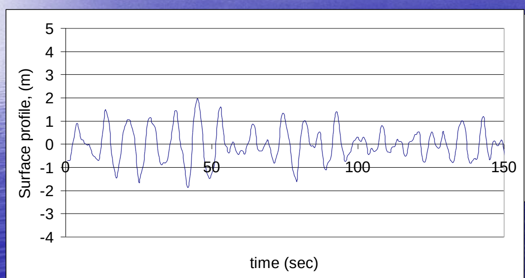
IRREGULARITY OF WAVES IN THE NATURE

Waves in the nature? *'CALM SEAS ONLY LIVE IN DREAMS'*



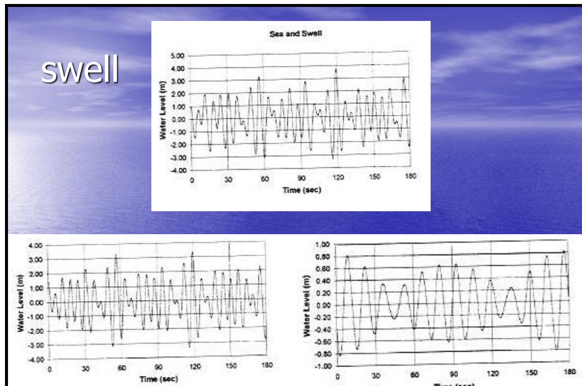
Sea Waves

Sea waves (wind waves)
- fetch, storm duration
- direction= wind direction
- irregular
- short crested=wave crests do not have a long extent but consists of short segments

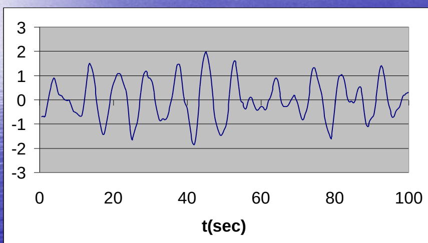
Sea Waves

Becomes swell when they move out of the generating area
- more orderly, almost regular
- long crested

swell



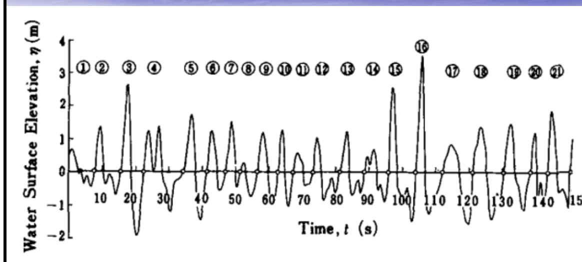
Definition of irregular wave parameters



Time series
Wave train

Zero up-crossing

Definition of irregular wave parameters



Characteristic wave parameters

- Hi, Ti
- Sort the waves according to Hi in descending order
- H1=Hmax
- Tmax is the period corresponding to Hmax
- Other characteristic wave parameters:

Hmean, Tmean
$H_{1/3}$ (Hs), $T_{1/3}$ (Ts)
$H_{1/10}$, $T_{1/10}$

$$H_{1/n} = \frac{n}{N}\sum_{i=1}^{N/n} Hi$$

$$T_{1/n} = \frac{n}{N}\sum_{i=1}^{N/n} Ti$$

i	Hi	Ti	Hi	Ti	Hmax	Tmax
1	5.13	7.63	**6.33**	**10.04**	**6.33**	**10.04**
2	3.59	7.01	**5.94**	**7.12**	**H1/10**	**T1/10**
3	5.94	7.12	**5.56**	**8.98**	**5.94**	**8.71**
4	0.75	4.92	**5.56**	**7.59**		
5	3.31	7.36	**5.53**	**8.25**		
6	3.69	7.32	**5.22**	**9.97**		
7	5.22	9.97	**5.21**	**8.78**		
8	5.53	8.25	**5.19**	**10.87**		
9	1.42	3.57	**5.13**	**7.63**	**H1/3=Hs**	**T1/3=Ts**
10	3.03	7.65	**4.95**	**10.16**	**5.46**	**7.94**
11	2.28	4.07	4.69	9.71		
12	2.99	7.25	4.62	8.24		
13	5.56	8.98	4.09	9.26		
14	3.71	7.96	3.79	5.72		
15	2.17	6.35	3.71	7.96		
16	0.81	3.40	3.69	7.32		
17	1.09	3.55	3.59	7.01		
18	3.79	5.72	3.31	7.36		
19	2.29	6.51	3.03	7.65		
20	3.00	7.67	3.00	7.67		
21	6.33	10.04	2.99	7.25		
22	4.09	9.26	2.29	6.51		
23	5.19	10.87	2.28	4.07		
24	5.21	8.78	2.17	6.35		
25	4.95	10.16	1.65	4.51		
26	5.56	7.59	1.43	5.43		
27	4.69	9.71	1.42	3.57		
28	1.43	5.43	1.09	3.55		
29	1.65	4.51	0.81	3.40		
30	4.62	8.24	0.75	4.92	**Hmean**	**Tmean**
					3.63	**7.23**

SPECTRUM OF SEA WAVES

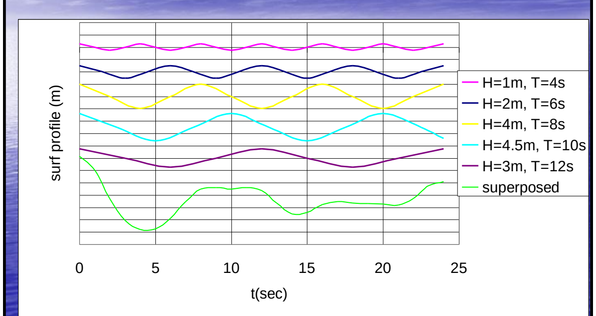
- The concept of spectrum….. Newton: Sunlight can be decomposed into a spectrum of colours with the aid of prism
- Spectrum indicates how the intensity of light varies with respect to its wavelength.
- CE side of view, spectrum indicates how the energy of the waves varies wrt its frequency and direction

SPECTRUM OF SEA WAVES

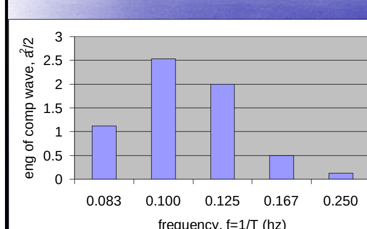
Sea waves (very random)= an infinite number of wavelets with different frequencies and directions

The distribution of the energy of these wavelets
wrt frequency= frequency spectrum
wrt both frequency and direction= directional spectrum

SPECTRUM OF SEA WAVES



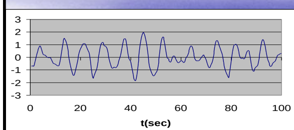
FREQUENCY SPECTRUM



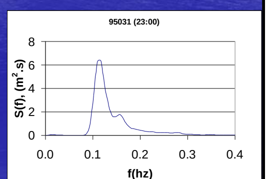
T(s)	f(hz)	a(m)	$E=a^2/2$
12	0.083	1.5	1.13
10	0.100	2.25	2.53
8	0.125	2	2.00
6	0.167	1	0.50
4	0.250	0.5	0.13

Spectral representation of superposed waves

FREQUENCY SPECTRUM

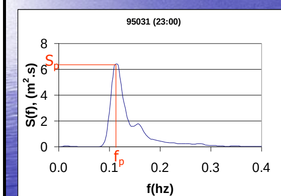


$$\eta_j = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi f_n t_j) + \sum_{n=1}^{\infty} b_n \sin(2\pi f_n t_j)$$



S(f)= frequency spectral density function (m^2/s)

Parameters derived from frequency spectrum



The moments of the wave spectrum

$$m_n = \int_{f=0}^{f=\infty} f^n S(f)\,df$$

The zero moment (n=0) is area under the spectrum

$$m_0 = \int_{f=0}^{f=\infty} S(f)\,df$$

$H_{m0}=4.004\sqrt{m_0} \approx Hs$

$T_p=1/f_p=1.05T_s$

$$T_2 = \sqrt{\frac{m_0}{m_2}} = T_{mean}$$

swell

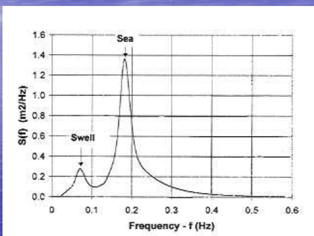


Figure 3.10 Wave Spectrum with Sea and Swell.